

1. b. $\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 6 & 0 & 1 \end{array} \right] \rightsquigarrow \text{R2-2R1} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array} \right] \text{ so } A^{-1} \text{ does not exist.}$

2. a. i. $\left[\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array} \right] \rightsquigarrow 2\text{R2} \left[\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 6 & 10 & 0 & 2 \end{array} \right] \rightsquigarrow \text{R2-3R1} \left[\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{array} \right] \rightsquigarrow \text{R1-3R2} \left[\begin{array}{cc|cc} 2 & 0 & 10 & -6 \\ 0 & 1 & -3 & 2 \end{array} \right] \rightsquigarrow$
 $\quad \quad \quad {}^{1/2}\text{R1} \left[\begin{array}{cc|cc} 1 & 0 & 5 & -3 \\ 0 & 1 & -3 & 2 \end{array} \right] \text{ So } A^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$

ii.

$$\begin{aligned} A^{-1}(A\mathbf{x}) &= A^{-1}\mathbf{b} \\ \mathbf{x} &= \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ \mathbf{x} &= \begin{bmatrix} 3 \\ -1 \end{bmatrix} \end{aligned}$$

iii. $\mathbf{b} = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$

c. i. $\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array} \right] \rightsquigarrow \begin{array}{l} \text{R1-R2} \\ \text{R3-R1} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array} \right] \rightsquigarrow \text{R2-R3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array} \right] \rightsquigarrow$
 $\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & -1 \end{array} \right]$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

ii.

$$\begin{aligned} A^{-1}(A\mathbf{x}) &= A^{-1}\mathbf{b} \\ \mathbf{x} &= \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix} \end{aligned}$$

iii.

$$\mathbf{b} = 3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

3. a. $(BAB^{-1})^2 = BAB^{-1}BAB^{-1} = BAAB^{-1} = BA^2B^{-1}$

b. [not assigned] $(BAB^{-1})^n = BA(B^{-1}B)A(B^{-1}B)A(B^{-1}B)AB^{-1} \dots BA(B^{-1}B)AB^{-1} = BA^nB^{-1}$

c. By proposition 3.4 $(BAB^{-1})^{-1} = (B^{-1})^{-1}(BA)^{-1} = B(A^{-1}B^{-1}) = BA^{-1}B^{-1}$ but we must assume that A and B are invertible.

Note that this means part (b) holds for negative integers as well.

4. A is invertible so there exists an A^{-1} . So then

$$\begin{aligned} A\mathbf{x} &= 7\mathbf{x} \\ A^{-1}A\mathbf{x} &= A^{-1}7\mathbf{x} \\ \mathbf{x} &= A^{-1}7\mathbf{x} \\ \frac{1}{7}\mathbf{x} &= A^{-1}\mathbf{x} \end{aligned}$$

$$\Rightarrow A^{-1}\mathbf{x} = \frac{1}{7}\mathbf{x}.$$